Nguyen Gia Hien Tu

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Education

Bachelor of Applied Science, Honours Computer Science Co-op **September 2019 – June 2024**
McMaster University, Hamilton, ON

- Graduated with ***Summa Cum Laude*** by achieving a **CGPA** of **3.97/4.0** (or 11.7 on a 12.0 scale)
- Honored on the **Deans' Honour List**

Ontario Secondary School Diploma **August 2018 – May 2019**
Columbia International College, Hamilton, ON

- Graduated with an average of **96%** in all six Grade 12 University Preparation courses
- Received the **Ontario Scholar** award for achieving an average of at least 80% in any six Grade 12 University Preparation courses

Skills

Programming Language: Python, SQL, Shell Scripting (Bash), Java, Haskell, Go, Javascript
Software: Git, GitHub, GitHub Actions, AWS, Docker, Kubernetes, OpenShift, Terraform, Snowflake, Pytest, FastAPI
Project Management Tool: Confluence, Jira, Notion

Experience

Catalyst Program Data and AI, Data and ML Engineer **May 2024 – Present**
Sanofi Toronto AI Center of Excellence

- Advanced skills in writing infrastructure as code (**IaC**) by implementing and enhancing **Terraform** templates to deploy **AWS** resources, such as **S3** bucket, **Lambda** function and **EMR**; as well as integrating AWS with **Informatica Cloud (IICS)**, **Snowflake** and **Apache Airflow**; enabling other technical teams to swiftly setup their AWS infrastructure
- Familiarized with developing automated pipelines through experimenting, supporting and improving pipeline to **process** and **extract unstructured and structured data** from PDF files using **S3** bucket, **Lambda** function and **AWS Textract** service in **Python**
- Demonstrated technical excellence by developing and thoroughly testing workflows that handle access requests from **IICS CDMP**, record the requests to **Snowflake** and configure cross-account **S3 Access Grants**
- Spearheaded the architecture design and end-to-end implementation of a secure, scalable and reliable **enterprise-grade** solution for deploying **containerized microservices** with automated test cases, including an **email notification** API service; leveraging **OpenShift** for container orchestration, **Docker** for containerization, **JFrog Artifactory** for artifact management, and an internal **API management platform**, in compliance with enterprise deployment standards, **CI/CD** and **DevOps** best practices, and security policies
- Improved researching and design thinking skills, and technical excellence by exploring, developing and deploying a production-ready enterprise chatbot for data asset discovery using **Generative AI**, **Large Language Models (LLMs)** and **Retrieval-Augmented Generation (RAG)** framework, implementing **REST APIs** using **FastAPI**, integrating **Snowflake Cortex** for vector search, orchestrating multiple **LLMs** to understand user queries and generate personalized responses, enhancing secure authentication and authorization using **OAuth 2.0** with **Microsoft Entra ID (Microsoft Azure AD)**, and deploying the **containerized** application via **OpenShift**, **JFrog**, **Docker** and **GitHub Actions**, in accordance with **DevOps** best practices
- Strengthened **cross-functional collaboration** skills by working closely with the **business team** to gather and analyze user requirements, and collaborating with other technical team members to design the system **architecture** and document the details in a comprehensive **design specification document**

Data Engineer Intern **May 2022 – August 2023**

Sanofi Toronto AI Center of Excellence

- Extensively broadened software development and design thinking skills through developing and maintaining an internal Data and AI platform using **FastAPI**, **Pydantic**, **SQLModel** and **microservices architecture** to enable hundreds of data scientists and MLOps engineers to research, develop and deploy AI models for healthcare purposes
- Actively learned and applied a wide range of the latest **cloud-native** technologies by developing the platform with Amazon Web Services (**AWS**), containerization platform (**Docker**), Kubernetes (**AWS EKS**), container registry (**AWS ECR**)
- Exposed to **DevOps** process by maintaining and improving **CI/CD pipelines** using **GitHub Actions**, **ArgoCD**, **Helm**, **Terraform** and shell scripting (**Bash**) following the **GitOps** approach
- Exhibited **documentation** and **mentorship** skills by creating onboarding guides for new users and developers, training incoming developers and co-op students, documenting new and existing features, and updating work progress through **Jira** tickets

Undergraduate Teaching Assistant **January 2022 – April 2022**

Automata and Computability - COMPSCI 2AC3

- Enhanced **interpersonal skills** through assisting students with the course materials using various communication channels, including Microsoft Teams, emails, weekly office hours, and providing documentation
- Demonstrated **excellent technical knowledge** by creating solutions for assignments, marking and providing feedback on student assignments, with more than **200** submissions graded
- Exhibited **time management** skills by attending weekly meetings with the instructor to report progress and discuss future work

Undergraduate Teaching Assistant **September 2021 – December 2021**

Introduction to Computational Thinking - COMPSCI 1JC3

- Developed **organization skills** by preparing and conducting tutorials to assist students with their knowledge in **Haskell**
- Improved **communication skills** through assisting students with the course materials using various communication channels, including Microsoft Teams, emails and weekly office hours
- Exhibited **time management** skills by attending weekly meetings with the instructor to report progress and discuss future work

Projects

DDO Vale Puzzle **July 2021 - August 2021**

- Developed a logic puzzle game with a variety of board sizes from 3x3 to 9x9 using **Java**, **Swing** and **Launch4j**
- Applied in-class and self-educated knowledge on **algebra (matrix multiplication)** and **object-oriented programming (OOP)** to create the game

Sorting Visualizer **February 2021 - March 2021**

- Developed an interactive sorting algorithm visualizer in **Python** and **Pygame** by applying in-class knowledge of **sorting algorithms** to animate and compare multiple sorting algorithms in real time
- Applied self-taught knowledge in **PyInstaller** to export the application as a standalone executable file

Extracurricular Activities

McMaster Start Coding - Volunteer and Facilitator **September 2019 – April 2022**

- Visited schools to introduce children from grades 1 to 8 to Computer Science design thinking and programming knowledge through creating images and games
- Utilized the club's **Elm**-based interactive tool to facilitate learning sessions and encourage problem-solving skills